

# Alejandro Sánchez Yalí

## AI Full Stack Engineer



✉ [asanchezyali@gmail.com](mailto:asanchezyali@gmail.com) | [in asanchezyali](#) | [asanchezyali](#) | [asanchezyali.com](#)

## SUMMARY

AI Full Stack Engineer with 8+ years designing and shipping web applications end to end and 3+ years building LLM-powered products. Architect and lead engineer of three deployed systems – an AI WhatsApp support platform, an AI-assisted recruitment platform and a surgical supply chain system – spanning Python/FastAPI backends, Next.js/TypeScript frontends, PostgreSQL and cloud deployment. M.Sc. in Mathematics and 11 years teaching mathematics and physics, which shows up in design docs and code review.

## TECHNICAL SKILLS

- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL, Rust (personal projects)
- **Frontend:** React.js (7+ years), Next.js (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, Effect, Three.js
- **Backend:** FastAPI, Django, Django REST Framework, Celery, Flask, NestJS, Node.js, Express, RESTful APIs, GraphQL, microservices
- **AI & LLMs:** LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, PyTorch, TensorFlow
- **Databases & Infra:** PostgreSQL, Redis, MongoDB, MySQL, Firebase, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Engineering:** Clean architecture, DDD, SOLID, TDD, Vitest, Jest, React Testing Library, code review

## EXPERIENCE

**Independent AI Consultant** Oct 2026 – Present  
*Freelance Remote*

- Lead engineer of Aluna, an AI recruitment platform for staffing agencies (see below)

**AI Specialist** Aug 2025 – Sep 2026  
*Lapzo Remote, Mexico*

- Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides
- Owned the architecture connecting LLM services, ElevenLabs voice synthesis and the real-time layer built with Next.js, NestJS and Firebase Realtime Database
- Closed 57 tasks and 6 epics in the first nine months of 2026 with a single production bug, per the team's board
- Built the AI content-generation pipelines (LangChain, LangGraph, n8n), with Celery workers running generation in the background

**Independent Software & AI Consultant** Aug 2024 – Jul 2025  
*Freelance Remote*

- Lead engineer of CREARIA's multi-tenant WhatsApp and email AI agent platform, in production with several client businesses including a real-estate agency: led a team of three across 5 FastAPI services and 2 Next.js dashboards on AWS ECS, with RAG over pgvector, multi-provider routing through LiteLLM, AI lead scoring and human escalation with full context; also led the AI coaching agent CREARIA delivers to one of its clients
- Delivered end-to-end web applications for business clients: React.js/Next.js/TypeScript on the front, Python/FastAPI/Django and Node.js/NestJS behind it, deployed on AWS with CI/CD and Stripe payments
- Ran technical design sessions with clients and made the technology decisions for each project; code reviewer for a client's development team

**Full Stack Engineer** Apr 2021 – Aug 2024  
*Monadical Remote, Canada/US*

- Full-stack engineer on ZoHuddle, a video conferencing app for scientific conferences (Neuromatch) that matches participants by mutual interests: React front end and Django back end, with Django Channels (WebSockets) and Twilio Video
- Full-stack engineer on OddSlings, an in-browser poker platform for desktop and mobile: React front end and Django back end on an event-stream architecture

- Built VC Tools, an LLM tool that retrieves and processes company information for investors (OpenAI, LangChain)
- Did code reviews on a distributed team across Canada, the US and Latin America

**Full Stack Engineer**  
BCFort

Aug 2018 – Oct 2020  
Medellín, Colombia

- Built blockchain dApp frontends (React, Web3.js) and analytics platforms (Django, Node.js) on AWS, GCP and IBM Cloud

**Professor of Mathematics & Physics (part-time)**  
Universidad de Antioquia

Jan 2010 – Oct 2021  
Medellín, Colombia

- Taught mathematics and physics; in the last year and a half also machine learning and data science with Python, NumPy, Pandas, TensorFlow and PyTorch
- Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research projects

## FEATURED PROJECTS

- **Plixiq – AI Customer Support on WhatsApp**  – Python, FastAPI, LiteLLM, PostgreSQL, Redis, Next.js, WebSocket
  - Architect and lead engineer of a multi-tenant SaaS where AI agents answer routine customer inquiries, book appointments and escalate the rest to human operators with full conversation context
  - Modular monolith (14 components plus a shared kernel) with boundaries enforced in CI; LiteLLM layer with a primary and fallback provider per agent, so an outage becomes a fallback
- **VitaStock – Surgical Supply Chain for Clinics**  – Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway
  - Architect and lead engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement; delivered, with rollout at the clinic pending
  - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Aluna – AI Recruitment Platform for Staffing Agencies**  – Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API
  - Lead engineer of an AI recruitment platform for staffing agencies – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
  - Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email

## OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion

## TALKS & WRITING

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024
- **Building Plixiq: a Multi-Tenant WhatsApp AI Support Platform** – architecture case study (2026)

## EDUCATION

**M.Sc. in Mathematics**  
Universidad de Antioquia

Aug 2010 - Sep 2013  
Medellín, Colombia

**B.Sc. in Teaching Mathematics and Physics**  
Universidad de Antioquia

Jan 2004 - Oct 2009  
Medellín, Colombia